Technical Note: Fine-Tuning Llama 3B for Emotional Intelligence

Core Danger: Catastrophic Forgetting

Issue: Training datasets sequentially in separate batches (e.g., Batch 1 then Batch 2) overwrites prior weights.
Result: The model completely loses the skills learned in earlier stages.

The Solution: Joint Training (Data Mixing)

Combine, downsample, and shuffle all specialized and general datasets into a single training run to teach the model empathy and conversational fluidity at the same time.

Step-by-Step Dataset Architecture

[Downsample Datasets] ➔ [Standardize to Llama Chat Template] ➔ [Shuffle All Data] ➔ [Single Run QLoRA]

1. Data Selection & Balancing (Target: ~45k–50k total samples)

Massively downsample huge instruction datasets so they do not drown out rare emotional data.

EmpatheticDialogues – Emotional Grounding – Use all ~25,000 samples.
ESConv – Support Strategies – Use all ~1,000–3,000 samples.
CounselChat – Expert Counseling – Use all ~1,000 samples.
DailyDialog – Multi-turn Flow – Downsample to 5,000 samples.
UltraChat 200K – Instruction Following – Downsample to 5,000 samples.
OpenAssistant (OASST1) – General Assistant – Downsample to 5,000 samples.

2. Standardization (Llama 3.2 JSON Format)

Convert all varying dataset structures (CSV, custom nested JSON, txt) into the strict, uniform Llama conversation structure:

```
[
{
  "messages": [
    {"role": "system", "value": "You are an emotionally intelligent AI assistant."},
    {"role": "user", "value": "I messed up a major presentation today."},
    {"role": "assistant", "value": "I hear you. Making a mistake in front of colleagues is incredibly stressful..."}
  ]
}
]
```

3. Shuffling and Single-Run Execution

Shuffle: Merge all 50,000 converted JSON objects into one array and randomize the order completely.

Hardware Tip: Use the Unsloth library with QLoRA (4-bit quantization).
VRAM: This workflow fits comfortably inside a single consumer GPU with 12GB–16GB VRAM and trains in a few hours.